

Entanglement generation from athermality

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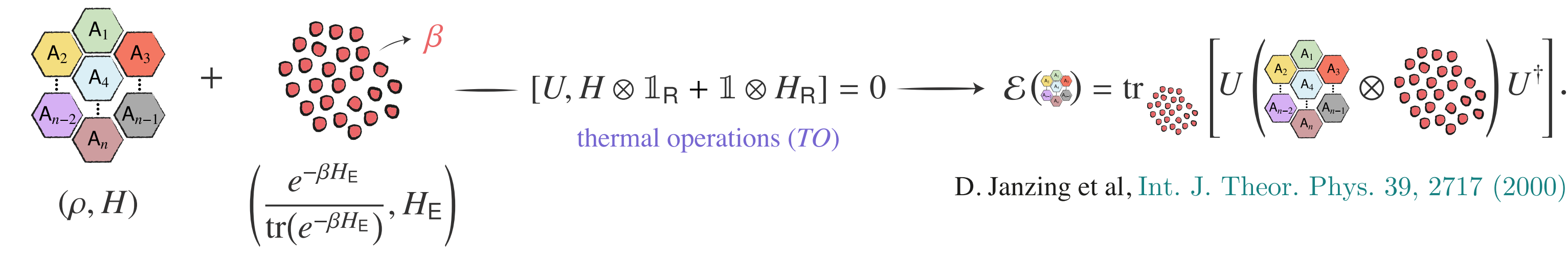
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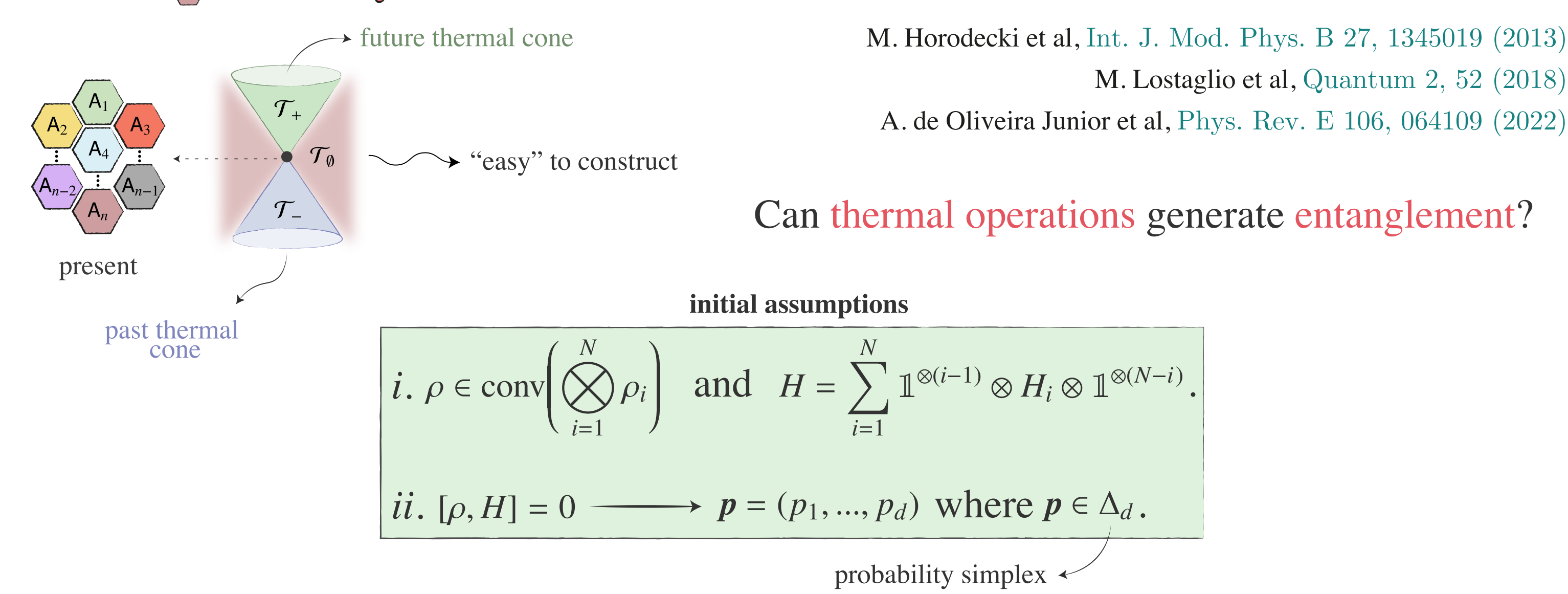
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What are the thermodynamic limitations for quantum correlations?

How to model thermodynamic transformations?

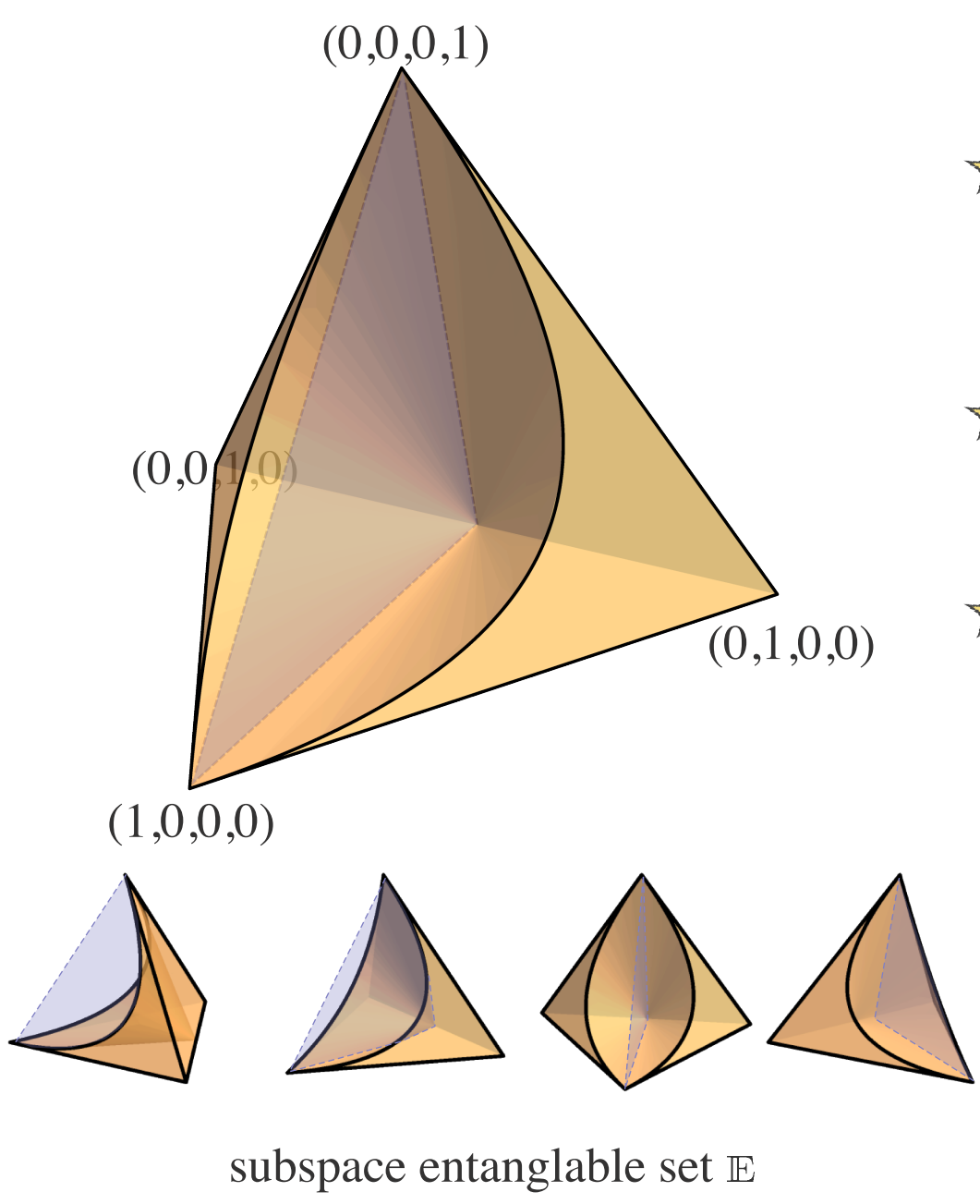


Given ρ and H , what is the set of states we can reach?



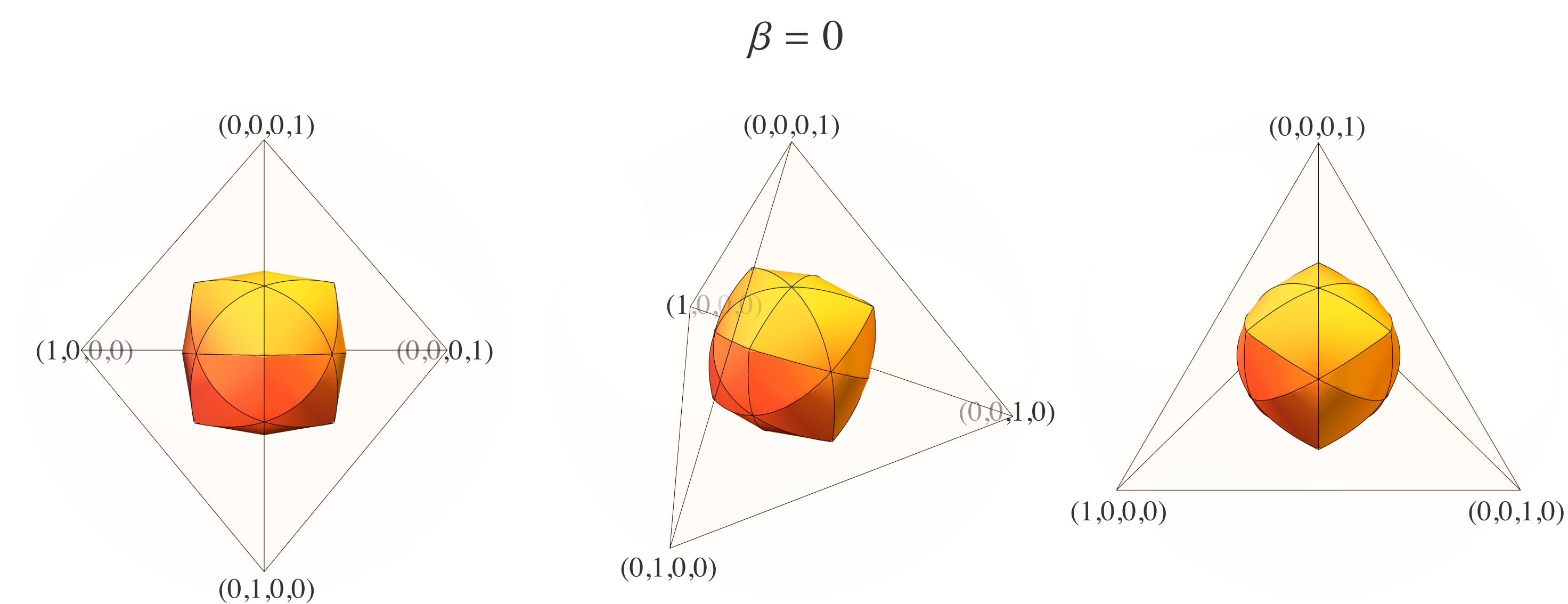
III. Two-qubit systems

Entanglement measure: $f(q) := 4q_1q_4 - (q_2 - q_3)^2$.

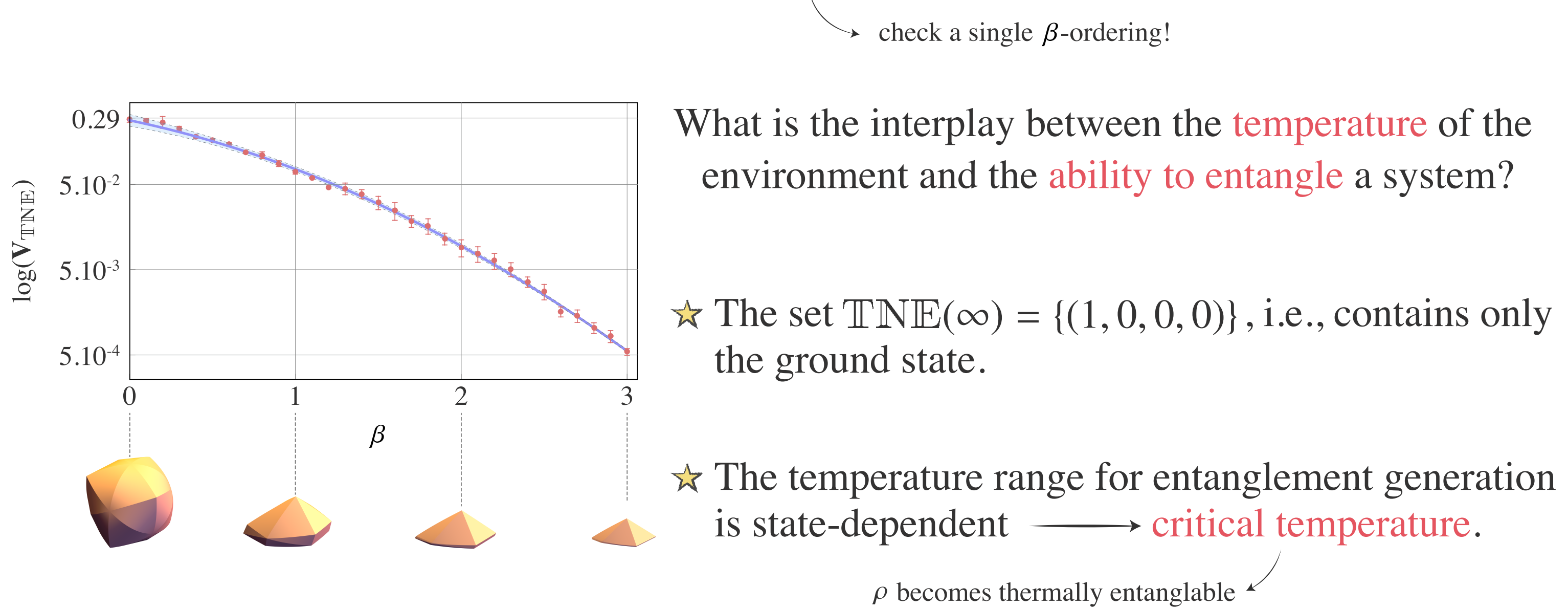


- ★ States in $\mathbb{E}$ can get entangled by some energy-preserving unitary of the form of the example!
- ★ The sets $\mathbb{NE}$ and $\mathbb{E}$ are independent of ambient temperature.
- ★ The normalised volume of $\mathbb{E}$ is approximately $V_{\mathbb{E}} \approx 21/62$.

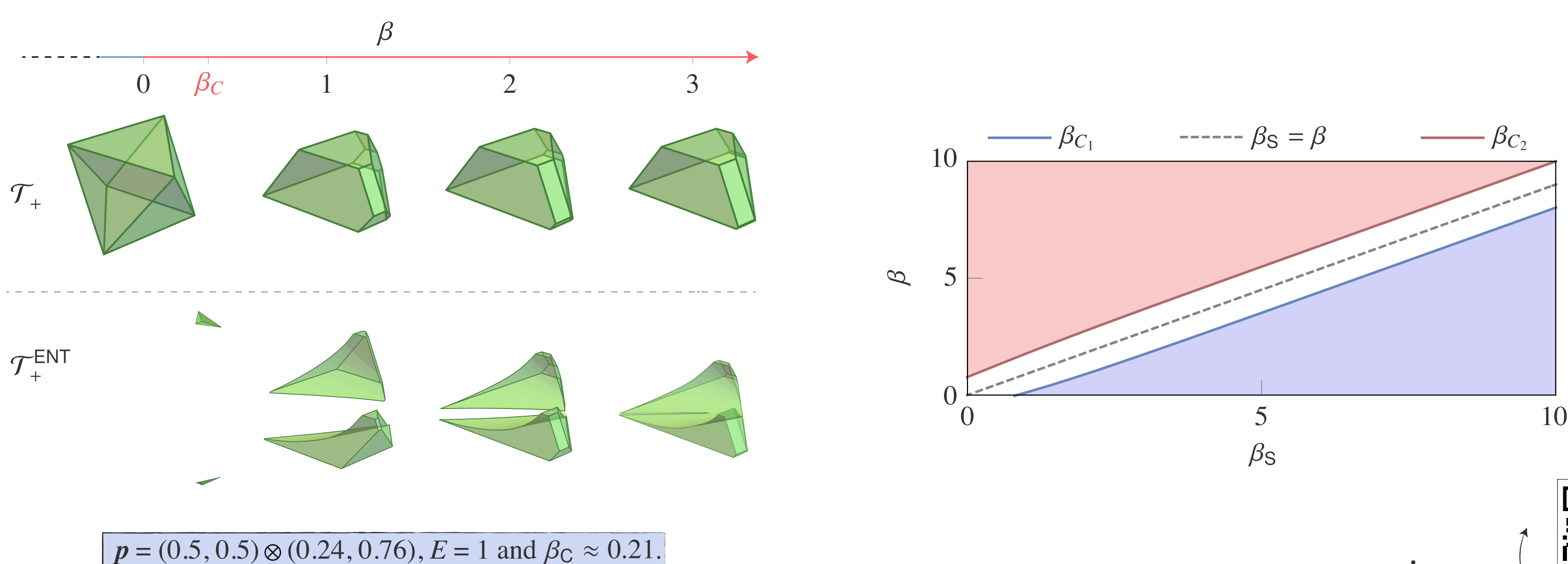
What is the set of states that cannot generate entanglement with the help of a thermal environment?



★ Analytical construction for $\beta = 0$ and a simple criterion for $\beta \neq 0$:



Regions of entanglement, thermal state & entanglability:



continue...

arXiv:2403.04842

Energy subspace unitaries:

$$\mathcal{U}_{\text{subs}} = \left\{ U_{\text{subs}} = \bigoplus_E U_E \right\}.$$

acts on $\mathcal{V}_E$ subspace of definite energy E

Example: two qubit systems ($H_A = H_B = E|1\rangle\langle 1|$).

$$H_{AB} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix} \longrightarrow U = \begin{pmatrix} \cos \theta & \sin \theta e^{i\phi} \\ -\sin \theta e^{-i\phi} & \cos \theta \end{pmatrix} \oplus \mathbb{1}_{\mathcal{H} \setminus \mathcal{V}_E}.$$

Entanglement measure $\mathcal{M} \longrightarrow f(p) := \min_{U \in \mathcal{U}_{\text{subs}}} \mathcal{M}(U \rho U^\dagger)$ and if $f(p) \geq 0$ then ρ is separable!

PPT criterion

Entanglement characterisation:

important definitions

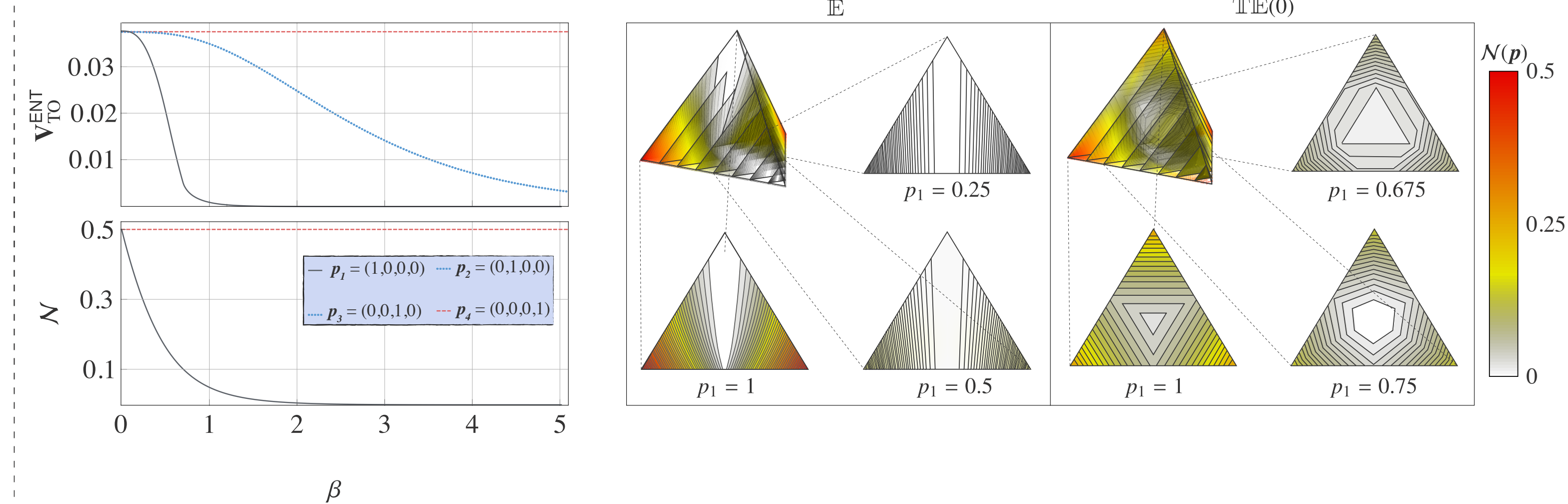
- Subspace non-entangleable set $\mathbb{NE} := \{p \mid f(p) \geq 0\}$ and subspace entangleable $\mathbb{E} := \Delta_d \setminus \mathbb{NE}$.
 - Future thermal cone of entanglement: $\mathcal{T}_+^{\text{ENT}}(p) := \{q \mid f(q) < 0, q \in \mathcal{T}_+(p)\} = \mathcal{T}_+(p) \cap \mathbb{E}$.
 - Thermally non-entangleable set: $\mathbb{TNE} = \{p \mid \forall \pi f(p_\pi) \geq 0\}$.
- extreme points of $\mathcal{T}_+(p)$

Entanglement quantification:

- $V_{\text{TO}}^{\text{ENT}}(p) := \frac{V[\mathcal{T}_+^{\text{ENT}}(p)]}{V(\Delta_d)}$, $V_{\mathbb{X}} := \frac{V(\mathbb{X})}{V(\Delta_d)}$, $\mathbb{X} = \mathbb{NE}, \mathbb{E}, \mathbb{TNE}$.
- $\mathcal{N}(\rho) = \frac{\|\rho^{T_A}\|_1 - 1}{2} = \sum_i \frac{|\lambda_i| - \lambda_i}{2}$.

To what extent can thermal operations produce entanglement, and how does this vary across different temperature regimes?

thermal volume of entanglement $V_{\text{TO}}^{\text{ENT}}(p)$ and maximum negativity $\mathcal{N}(\rho)$:



★ Maximum negativity is only reachable for very specific initial states!

★ ... but one can still obtain an amount of negativity close to the maximum.

IV. Generation of entanglement in the presence of dissipation

System undergoing an open dynamics:

$$\frac{d\rho(t)}{dt} = -i[H, \rho(t)] + \mathcal{L}_i[\rho(t)] \quad \text{where} \quad \mathcal{L}_i(\rho) = \sum_i r_i(t) \left[L_i(t) \rho L_i(t)^\dagger - \frac{1}{2} \{L_i(t)^\dagger L_i(t), \rho\} \right].$$

A. Kossakowski, *Rep. Math. Phys.* 3, 247 (1972)

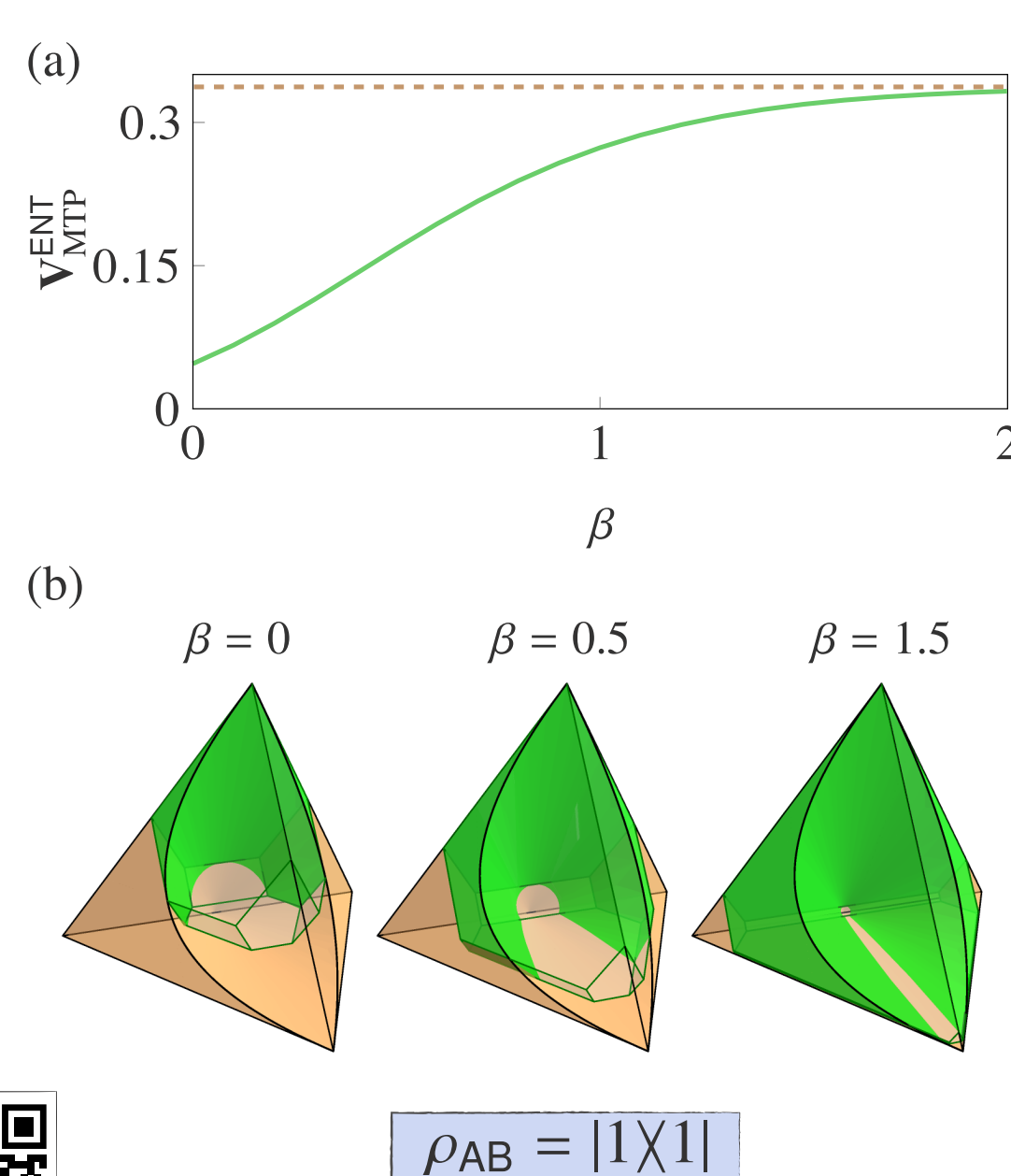
V. Gorini et al, *J. Math. Phys.* 17, 821 (1976)

G. Lindblad, *Commun. Math. Phys.* 48, 119 (1976)

Properties

- Stationary thermal state $\mathcal{L}_i(\rho) = 0$. $\xrightarrow{\text{Hi}}$ Markovian thermal process (MTP)
 - Covariance $\mathcal{L}_i[\mathcal{H}(\rho)] = \mathcal{H}[\mathcal{L}_i(\rho)]$. $\xrightarrow{-i[H, \rho]}$ state transformation conditions are known!
- M. Lostaglio et al, *Phys. Rev. A* 106, 012426 (2022)
- K. Korzekwa et al, *Phys. Rev. Lett.* 129, 040602 (2022)

Entanglement measure remains valid!



- ★ The difference between the entanglement generation via TO and MTP is most pronounced for small β .
- ★ For $\beta \rightarrow \infty$ MTP are as capable as TO for generating entanglement.

V. Remarks on higher-dimensional systems

- ★ Different local energy structures give rise to different entanglement structures.
- ★ Qubit-qutrit/Qubit-qudit chain of conditions.

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